

Applying Supervised Learning to Predict Student Dropout

by

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Table of contents:

1. <i>Approach Overview</i>	3
2. <i>Description of the Analysis</i>	3
2.1 <i>Stage 1: Applicant and Course Information</i>	3-5
2.2 <i>Stage 2: Student and Engagement Data</i>	5
2.3 <i>Stage 3: Academic Performance Data</i>	5-6
3. <i>Model Comparison and Risk Stages</i>	6
3.1 <i>Performance Change</i>	6
3.1.1 <i>Stage 1</i>	6-8
3.1.2 <i>Stage 2</i>	8-10
3.1.3 <i>Stage 3</i>	10-11
3.2 <i>Key Findings and Reflections</i>	11
4. <i>Business Impact, Insights and Visualisations</i>	11
4.1 <i>Inferences</i>	12-13
4.2 <i>Implications</i>	13

1. Approach Overview

The objective of this project was to develop supervised learning models to predict student dropout using institutional data, enabling earlier intervention and improved retention. A staged approach reflected data availability:

- Stage 1, applicant and course data
- Stage 2, engagement data
- Stage 3, academic performance data.

XGBoost and neural networks were trained at each stage to compare predictive performance across the student journey.

2. Description of the Analysis

2.1 Stage 1: Applicant and Course Information

Exploratory analysis of dataset structure, types, shape, duplicates, missing values, cardinality.

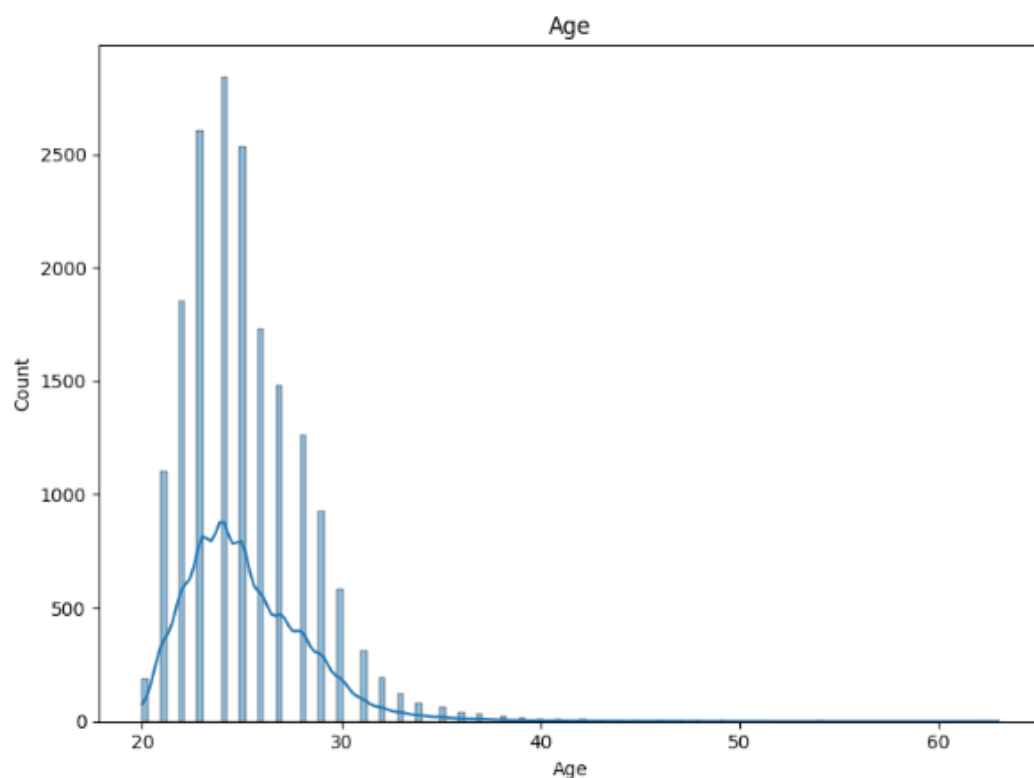


Figure 1. Stage 1: Histogram on numerical columns

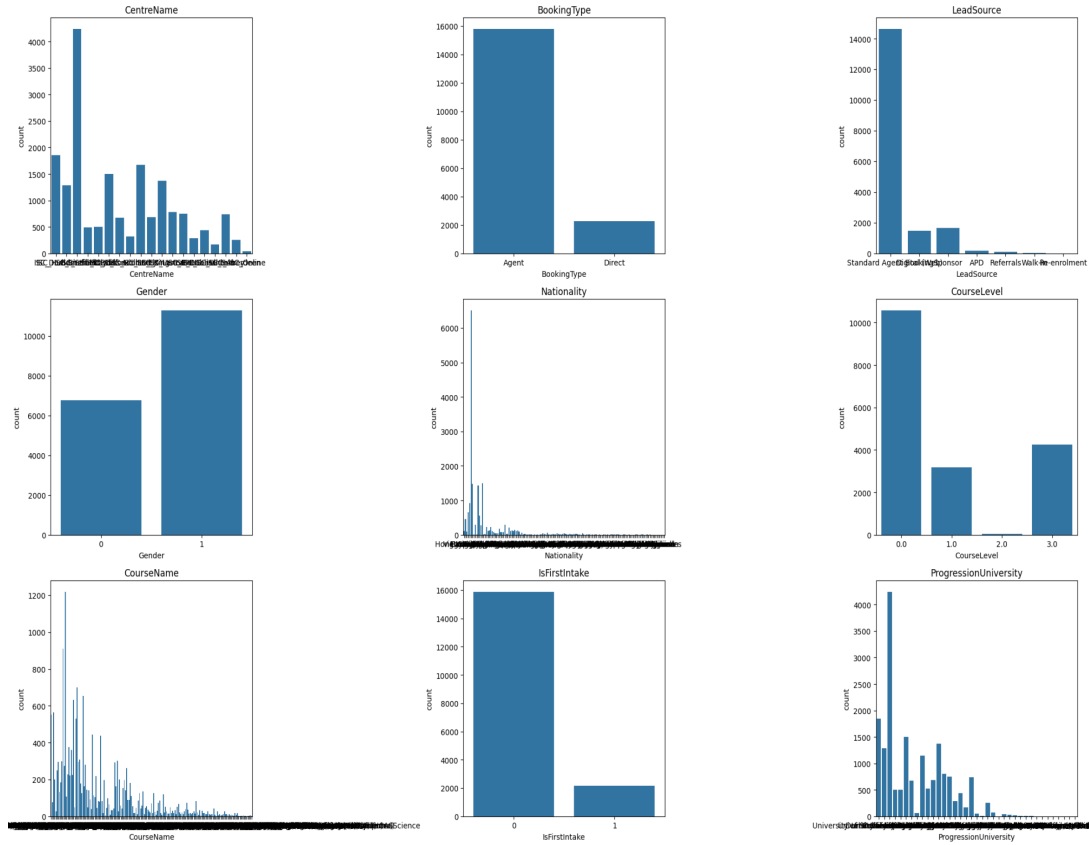


Figure 2. Stage 1: Bar charts on categorical columns

Preprocessing included engineering DateofBirth to Age, removing non-informative identifiers, excluding high-cardinality variables (>200 unique values), and dropping columns with $>50\%$ missing values. Data was split into training, validation, and test sets before encoding to prevent leakage.

Two models were implemented, XGBoost and neural network:

- Baseline model (V_1) was evaluated on the test set.
- Hyperparameter tuning was performed on the validation set ($V_2 - V_n$).
- The best configuration was re-evaluated on the test set.

Additionally, AUC was used due to class imbalance (85% positive, 15% negative).

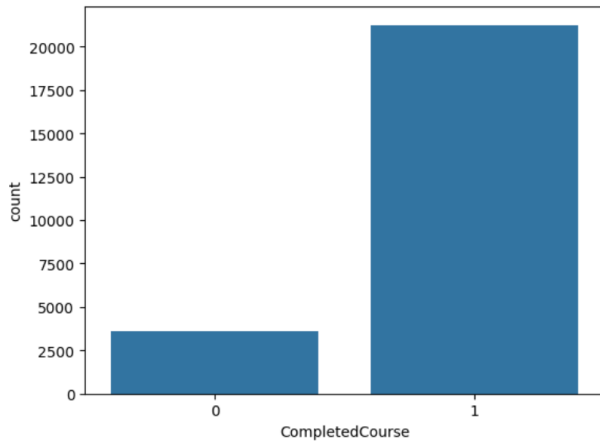


Figure 3. Stage 1: Imbalance on target variable

XGBoost interpretability was analysed using feature importance and SHAP, while neural network was evaluated using loss and AUC curves.

2.2 Stage 2: Student and Engagement Data

The same exploratory analysis and preprocessing pipeline were applied. AuthorisedAbsenceCount contained <2% missing values (0.83%), hence missing rows were removed.

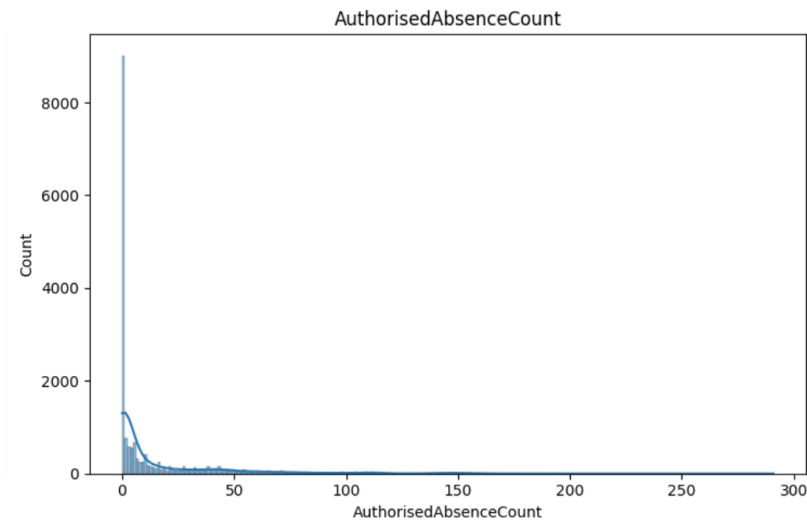


Figure 4. Stage 2: Histogram on AuthorisedAbsenceCount

The methodology remained consistent with Stage 1, using the same baseline hyperparameters and training, tuning, and evaluation framework to enable direct comparison.

2.3 Stage 3: Academic Performance Data

Stage 3 introduced academic performance variables, including “FailedModules”, “AssessedModules”, and “PassedModules”. The exploratory analysis and preprocessing pipeline remained consistent with Stage 2.

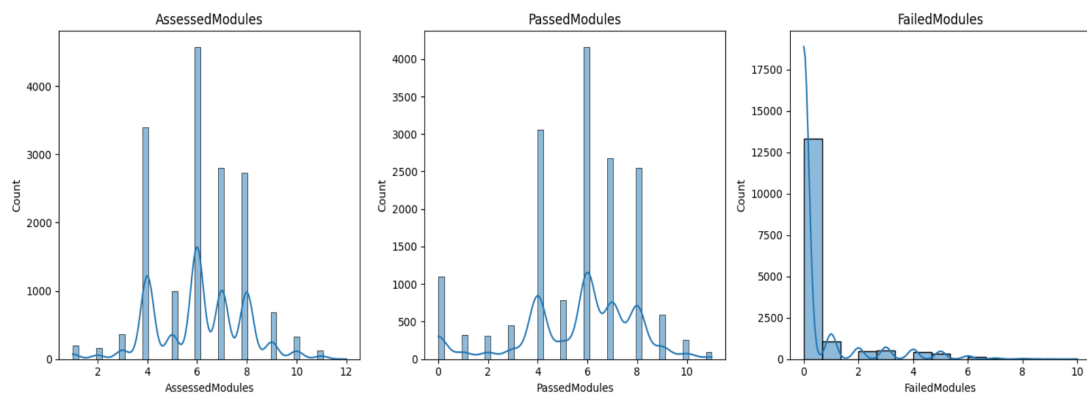


Figure 5. Stage 3: Histogram on AssessedModules, PassedModules, and FailedModules

These variables contained higher missing values (8.9%) and required imputation:

- AssessedModules, imputed with median

- PassedModules, imputed with mean
- FailedModules, imputed with median

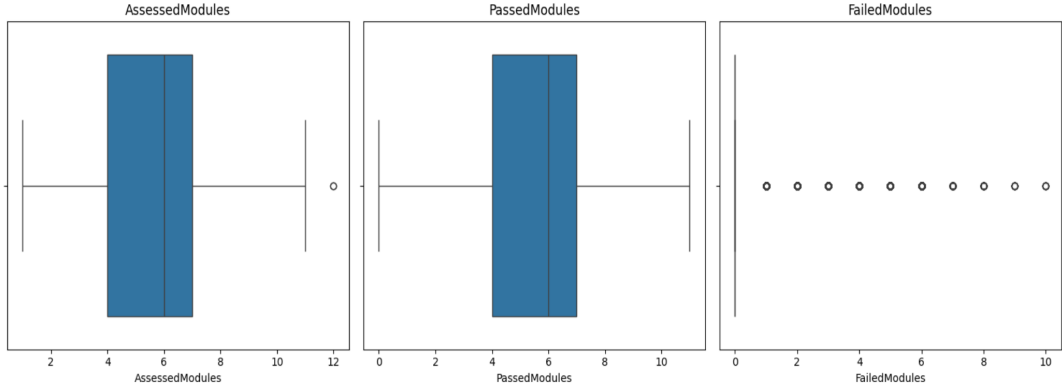


Figure 6. Stage 3: Boxplots on AssessedModules, PassedModules, and FailedModules

Stage 3 followed the same framework as Stage 2, using identical baseline hyperparameters for direct comparison before applying hyperparameter tuning.

3. Model Comparison and Risk Stages

3.1 Performance Change

The following aliases are used for hyperparameters and evaluation metrics:

Hyperparameter/Metric	Alias
n_estimators	n_e
learning_rate	l_r
max_depth	m_d
optimizer	opt
batch_size	b_s
hidden_layers	h_l
epochs	ep
units	un
Accuracy	A
Precision	P
Recall	R
Loss	L

Table 1. Hyperparameter/metric alias mapping

3.1.1 Stage 1

XGBoost (V_1) baseline model evaluated on test set:

Stage	Model	n_e	l_r	m_d	AUC	A	P	R	TN	FP	FN	TP
1	V_1	100	0.1	3	0.874	0.889	0.913	0.962	359	392	162	4099

Table 2. XGBoost Stage 1 (V_1) baseline model performance

XGBoost ($V_2 - V_6$) hyperparameter-tuned models on validation set:

Stage	Model	n_e	l_r	m_d	AUC	A	P	R	TN	FP	FN	TP
1	V_2	200	0.05	3	0.890	0.899	0.915	0.971	146	154	49	1656
1	V_3	200	0.05	5	0.900	0.899	0.916	0.970	149	151	51	1654
1	V_4	1500	0.01	6	0.901	0.899	0.918	0.967	153	147	56	1649
1	V_5	3000	0.01	6	0.900	0.899	0.920	0.965	157	143	59	1646
1	V_6	5000	0.005	4	0.899	0.900	0.917	0.970	151	149	52	1653

Table 3. XGBoost Stage 1 ($V_2 - V_6$) tuned models' performance

Best-tuned model (V_4) vs baseline (V_1) on test set:

Stage	Model	n_e	l_r	m_d	AUC	A	P	R	TN	FP	FN	TP
1	V_1	100	0.1	3	0.874	0.889	0.913	0.962	359	392	162	4099
1	V_4	1500	0.01	6	0.880	0.895	0.918	0.962	386	365	162	4099

Table 4. XGBoost Stage 1 (V_1 vs V_4) baseline vs best-tuned model

The baseline model (V_1) achieved a test AUC of 0.874. V_4 reached 0.901 on validation and 0.880 on the test set, indicating a modest improvement but suggesting mild overfitting.

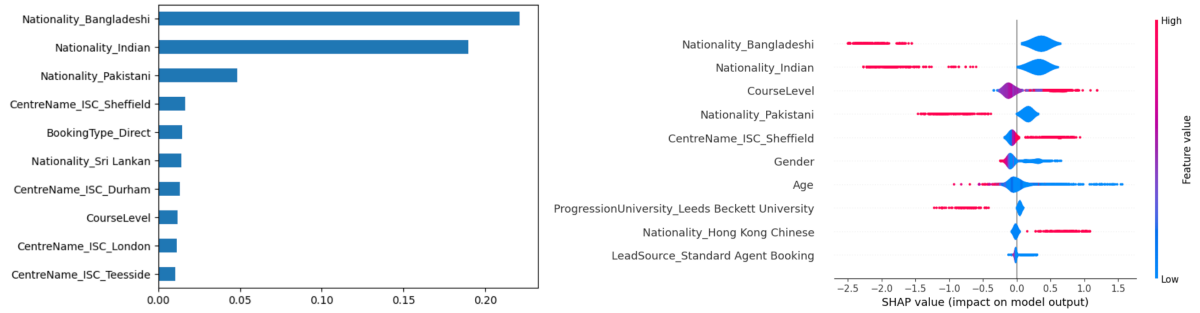


Figure 7. XGBoost Stage 1: Feature importance (left) and SHAP violin chart (right)

Nationality dominates both plots, with CourseLevel also important, SHAP highlights some features absent from feature importance, and vice versa.

Neural network (V_1) baseline model evaluated on test set:

Stage	Model	ep	opt	b_s	h_l	un	AUC	L	A	P	R	TN	FP	FN	TP
1	V_1	10	adam	32	1	10	0.874	0.278	0.891	0.913	0.964	361	390	154	4107

Table 5. Neural network Stage 1 (V_1) baseline model performance

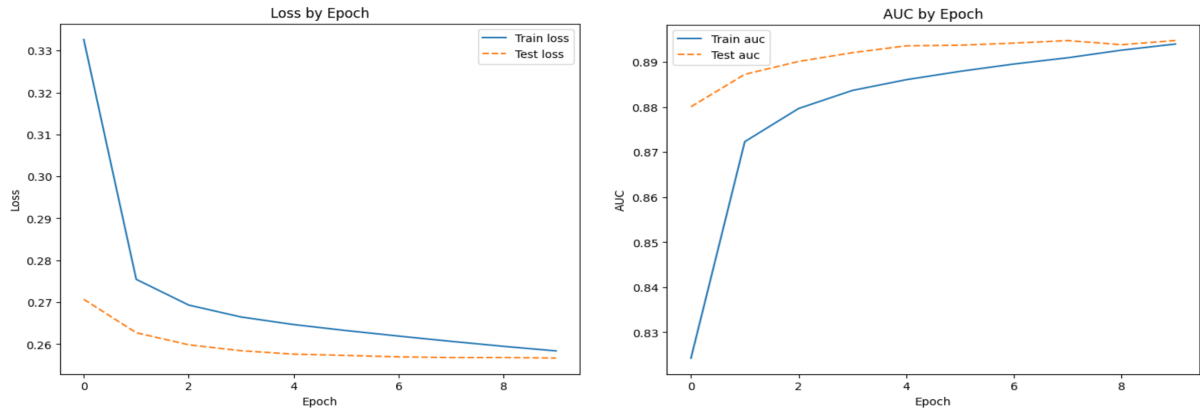


Figure 8. Neural network Stage 1: (V_1) baseline model Loss & AUC by Epoch

Neural network ($V_2 - V_6$) hyperparameter-tuned models on validation set:

Stage	Model	ep	opt	b_s	h_l	un	AUC	L	A	P	R	TN	FP	FN	TP
1	V_2	10	rmsprop	32	1	10	0.892	0.258	0.898	0.916	0.968	149	151	54	1651
1	V_3	15	adam	32	2	15	0.889	0.263	0.900	0.916	0.971	149	151	50	1655
1	V_4	10	adam	16	1	10	0.897	0.254	0.897	0.912	0.972	141	159	47	1658
1	V_5	15	adam	16	1	10	0.892	0.261	0.900	0.915	0.973	146	154	46	1659
1	V_6	10	adam	16	1	5	0.895	0.258	0.894	0.913	0.968	142	158	54	1651

Table 6. Neural network Stage 1 ($V_2 - V_6$) tuned models' performance

Best-tuned model (V_4) vs baseline (V_1) on test set:

Stage	Model	ep	opt	b_s	h_l	un	AUC	L	A	P	R	TN	FP	FN	TP
1	V_1	10	adam	32	1	10	0.874	0.278	0.891	0.913	0.964	361	390	154	4107
1	V_4	10	adam	16	1	10	0.873	0.281	0.893	0.913	0.966	361	390	147	4114

Table 7. Neural network Stage 1 (V_1 vs V_4) baseline vs best-tuned model

The baseline model (V_1) achieved a test AUC of 0.874. V_4 reached 0.897 on validation but 0.873 on the test set, suggesting slight overfitting. Overall, the best-performing **Stage 1 model was XGBoost (V_4)**, achieving a test AUC of **0.880**.

3.1.2 Stage 2

XGBoost Stage 1 vs 2 (V_1) baseline model evaluated on test set:

Stage	Model	n_e	l_r	m_d	AUC	A	P	R	TN	FP	FN	TP
1	V_1	100	0.1	3	0.874	0.889	0.913	0.962	359	392	162	4099
2	V_1	100	0.1	3	0.888	0.895	0.918	0.963	356	366	156	4093

Table 8. XGBoost Stage 1 vs 2 (V_1) baseline model

Model using identical hyperparameters improved from Stage 1 (0.874) to Stage 2 (0.888). This improvement is attributed to the additional feature AuthorisedAbsenceCount.

XGBoost ($V_2 - V_6$) hyperparameter-tuned models on validation set:

Stage	Model	n_e	l_r	m_d	AUC	A	P	R	TN	FP	FN	TP
2	V_2	500	0.05	3	0.920	0.908	0.928	0.967	162	127	56	1643
2	V_3	500	0.05	5	0.921	0.910	0.931	0.967	167	122	56	1643
2	V_4	1000	0.01	5	0.922	0.912	0.930	0.971	164	125	50	1649
2	V_5	10000	0.01	5	0.918	0.908	0.931	0.964	168	121	62	1637
2	V_6	5000	0.01	6	0.920	0.908	0.931	0.965	167	122	60	1639

Table 9. XGBoost Stage 2 ($V_2 - V_6$) tuned models' performance

Stage	Model	n_e	l_r	m_d	AUC	A	P	R	TN	FP	FN	TP
2	V_1	100	0.1	3	0.888	0.895	0.918	0.963	356	366	156	4093
2	V_4	1000	0.01	5	0.894	0.897	0.922	0.961	375	347	165	4084

Table 10. XGBoost Stage 2 (V_1 vs V_4) baseline vs best-tuned model

The baseline model (V_1) achieved a test AUC of 0.888. After tuning, V_4 reached 0.922 on validation and 0.894 on the test set, indicating modest improvement but mild validation overfitting.

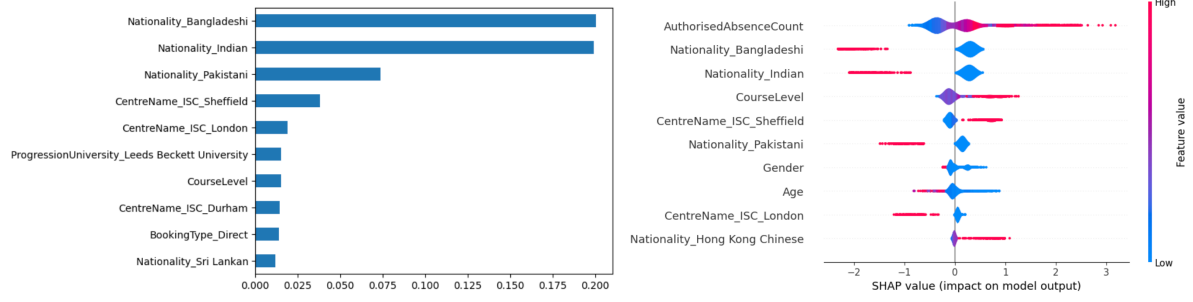


Figure 11. XGBoost Stage 2: Feature importance (left) and SHAP violin chart (right)

Feature importance ranks nationality highest, while SHAP shows AuthorisedAbsenceCount as most impactful.

Neural network Stage 1 vs 2 (V_1) baseline model evaluated on test set:

Stage	Model	ep	opt	b_s	h_l	un	AUC	L	A	P	R	TN	FP	FN	TP
1	V_1	10	adam	32	1	10	0.874	0.278	0.891	0.913	0.964	361	390	154	4107
2	V_1	10	adam	32	1	10	0.879	0.268	0.897	0.922	0.922	379	343	169	4080

Table 11. Neural network Stage 1 vs 2 (V_1) baseline model

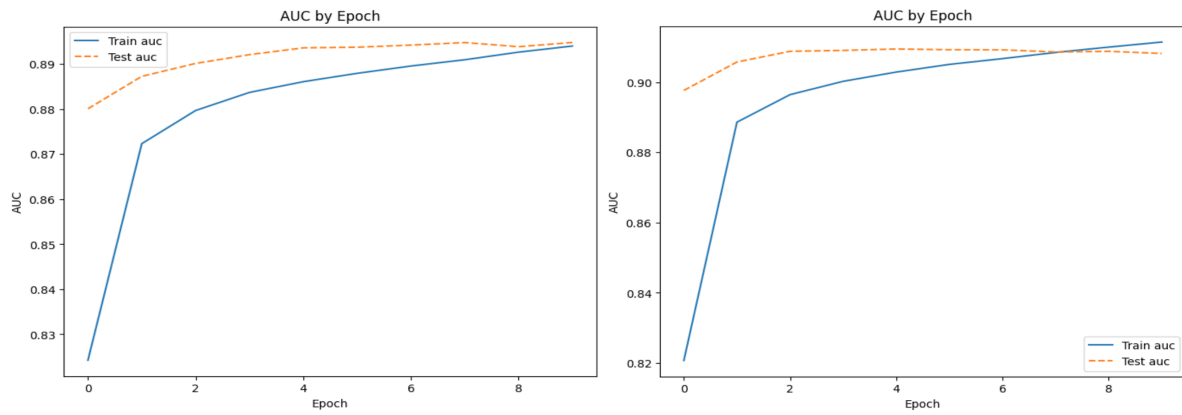


Figure 12. Neural network Stage 1 (left) vs 2 (right) AUC by Epoch

Stage 2 introduced AuthorisedAbsenceCount. While keeping the neural network architecture unchanged, AUC increased from 0.874 to 0.879.

Neural network (V_2 - V_6) hyperparameter-tuned models on validation set:

Stage	Model	ep	opt	b_s	h_l	un	AUC	L	A	P	R	TN	FP	FN	TP
2	V_2	10	rmsprop	32	1	10	0.905	0.241	0.905	0.928	0.964	161	128	61	1638
2	V_3	20	rmsprop	16	1	10	0.902	0.245	0.905	0.924	0.969	154	135	53	1646
2	V_4	15	adam	16	1	10	0.907	0.242	0.902	0.931	0.956	168	121	74	1625
2	V_5	15	adam	16	2	10	0.899	0.251	0.898	0.928	0.955	164	125	77	1622
2	V_6	15	adam	16	1	15	0.904	0.245	0.905	0.928	0.962	162	127	65	1634

Table 12. Neural network Stage 2 (V_2 - V_6) tuned models' performance

Stage	Model	ep	opt	b_s	h_l	un	AUC	L	A	P	R	TN	FP	FN	TP
2	V_1	10	adam	32	1	10	0.879	0.268	0.897	0.922	0.922	379	343	169	4080
2	V_4	15	adam	16	1	10	0.875	0.277	0.897	0.925	0.956	395	327	187	4062

Table 13. Neural network Stage 2 (V_1 vs V_4) baseline vs best-tuned model

The baseline model (V_1) achieved a test AUC of 0.879. V_4 reached 0.907 on validation but 0.875 on the test set, indicating mild overfitting. Overall, the best Stage 2 model was **XGBoost model (V_4)**, achieving a test AUC of **0.894**.

3.1.3 Stage 3

XGBoost Stage 2 vs 3 (V_1) baseline model evaluated on test set:

Stage	Model	n_e	l_r	m_d	AUC	A	P	R	TN	FP	FN	TP
2	V_1	100	0.1	3	0.888	0.895	0.918	0.963	356	366	156	4093
3	V_1	100	0.1	3	0.9986	0.987	0.988	0.997	671	51	13	4236

Table 14. XGBoost Stage 2 vs 3 (V_1) baseline model

With identical hyperparameters, Stage 3 achieved an AUC of 0.999 vs 0.888 in Stage 2, likely driven by AssessedModules, FailedModules, and PassedModules.

XGBoost ($V_2 - V_4$) hyperparameter-tuned models on validation set:

Stage	Model	n_e	l_r	m_d	AUC	A	P	R	TN	FP	FN	TP
3	V_2	200	0.05	3	0.9980	0.987	0.989	0.996	271	18	7	1692
3	V_3	200	0.05	5	0.9986	0.987	0.990	0.995	272	17	8	1691
3	V_4	400	0.05	5	0.9990	0.989	0.992	0.995	275	14	8	1691

Table 15. XGBoost Stage 3 ($V_2 - V_4$) tuned models' performance

Stage	Model	n_e	l_r	m_d	AUC	A	P	R	TN	FP	FN	TP
3	V_1	100	0.1	3	0.9986	0.987	0.988	0.997	671	51	13	42
3	V_4	400	0.05	5	0.9992	0.991	0.993	0.997	692	30	14	42

Table 16. XGBoost Stage 3 (V_1 vs V_4) baseline vs best-tuned model

The baseline model (V_1) achieved a test AUC of 0.9986. V_4 reached 0.9990 on validation set and 0.9992 on test set, indicating strong predictive signal with virtually no generalisation gap.

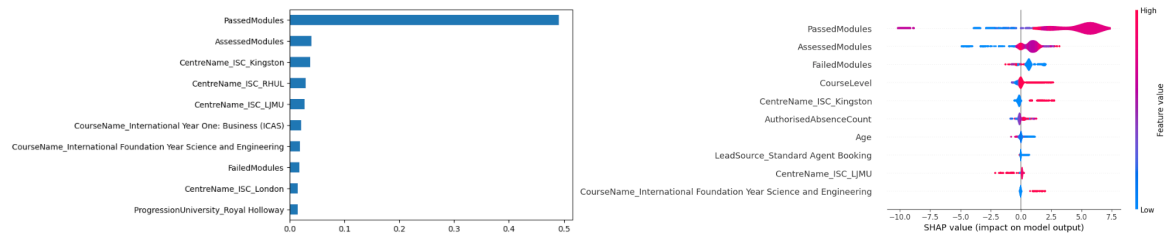


Figure 13. XGBoost Stage 3: Feature importance (left) and SHAP violin chart (right)

Both methods identify PassedModules as the dominant feature, explaining the near-perfect AUC.

Neural network Stage 2 vs 3 (V_1) baseline model evaluated on test set:

Stage	Model	ep	opt	b_s	h_l	un	AUC	L	A	P	R	TN	FP	FN	TP
2	V_1	10	adam	32	1	10	0.879	0.268	0.897	0.922	0.922	379	343	169	4080
3	V_1	10	adam	32	1	10	0.982	0.102	0.967	0.968	0.994	584	138	24	4225

Table 17. Neural network Stage 2 vs 3 (V_1) baseline model

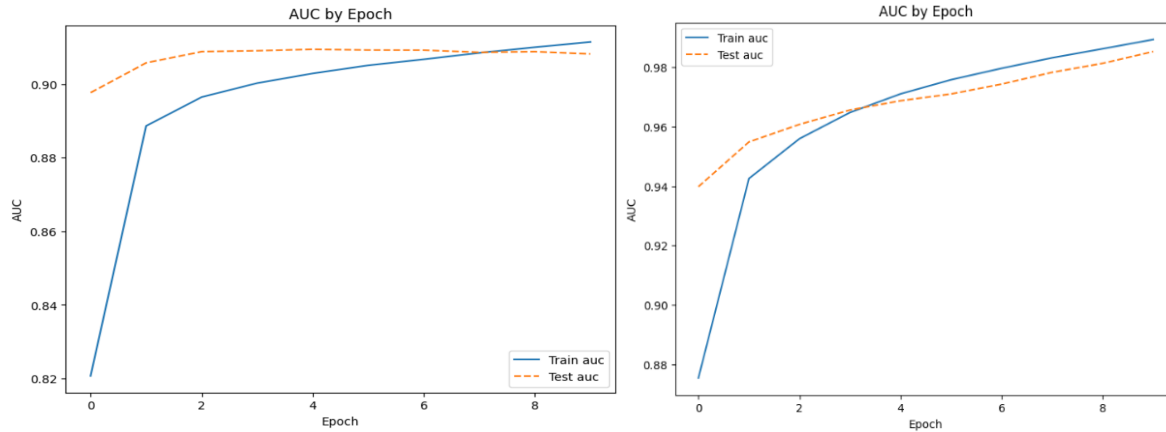


Figure 14. Neural network Stage 2 (left) vs 3 (right) AUC by Epoch

AUC increased from 0.879 to 0.982 in Stage 3, after adding academic performance features.

Neural network ($V_2 - V_6$) hyperparameter-tuned models on validation set:

Stage	Model	ep	opt	b_s	h_l	un	AUC	L	A	P	R	TN	FP	FN	TP
3	V_2	10	rmsprop	32	1	10	0.973	0.132	0.960	0.969	0.984	236	53	26	1673
3	V_3	15	adam	64	1	10	0.983	0.096	0.972	0.974	0.994	244	45	11	1688
3	V_4	15	adam	64	2	10	0.997	0.040	0.988	0.989	0.997	270	19	5	1694
3	V_5	15	adam	64	1	15	0.992	0.075	0.977	0.978	0.995	251	38	8	1691

Table 18. Neural network Stage 3 ($V_2 - V_5$) tuned models' performance

Stage	Model	ep	opt	b_s	h_l	un	AUC	L	A	P	R	TN	FP	FN	TP
3	V_1	10	adam	32	1	10	0.982	0.102	0.967	0.968	0.994	584	138	24	4225
3	V_4	15	adam	64	2	10	0.990	0.067	0.979	0.978	0.998	626	968	4241	

Table 19. Neural network Stage 3 (V_1 vs V_4) baseline vs best-tuned model

In summary, across all models evaluated on the Stage 3 test dataset, including both XGBoost and neural network approaches, the **XGBoost model (V_4)** performed best, achieving an AUC of **0.9992**.

3.2 Key Findings and Reflections

Model performance improved as more student information became available across the three stages. In Stage 1, using only applicant and course information, the best model achieved an AUC of 0.880, enabling early identification of at-risk students.

In Stage 2, adding engagement data increased AUC to 0.894, suggesting behavioural indicators provide stronger signals of dropout risk while still allowing relatively early intervention. The largest improvement occurred in Stage 3, where academic performance variables increased AUC to 0.999. Although these variables provide the strongest predictive signal, predictions occur later in the student journey, reducing the intervention window.

4. Business Impact, Insights & Visualisations

4.1 Inferences

Predictive performance improves as more student data becomes available, with Stage 3 providing the strongest signal, though Stages 1–2 still offer early indicators of dropout risk.

Stage	Best Model Name	Predictive Performance (AUC)
1 (Early Stage)	XGBoost_Stage1_v4	0.880
2 (Mid Stage)	XGBoost_Stage2_v4	0.894
3 (Late Stage)	XGBoost_Stage3_v4	0.999

Table 20. Comparative analysis of the best-performing models from Stages 1, 2, and 3

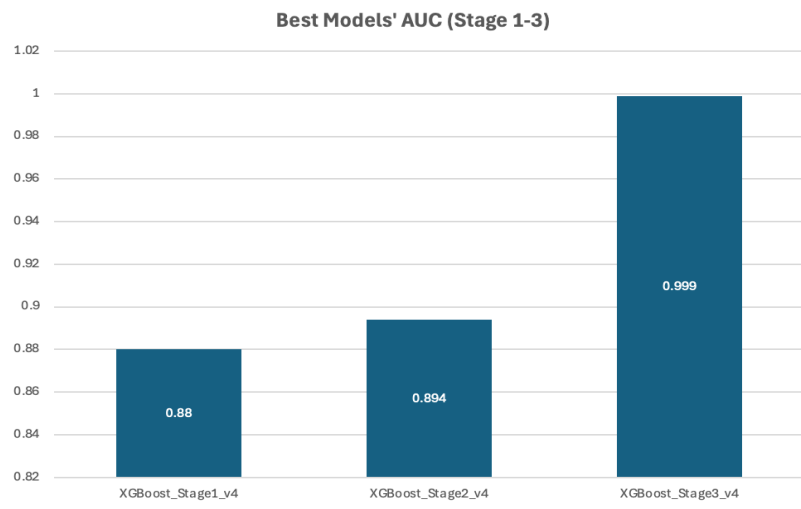


Figure 15. Best Models' AUC (Stage 1- 3)

SHAP analysis shows that newly introduced variables increasingly dominate model predictions, indicating their growing importance in the model. In particular, Stage 3 academic performance features emerge among the most influential predictors.

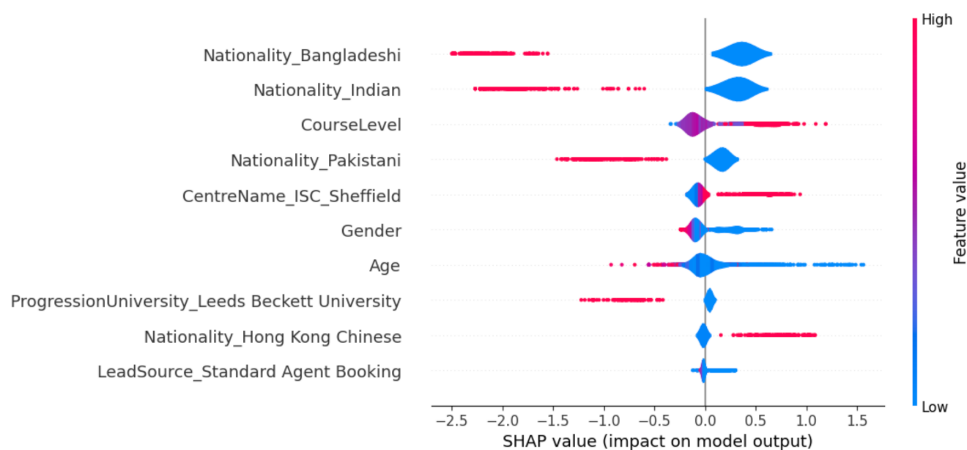


Figure 15. XGBoost Stage 1 SHAP violin chart

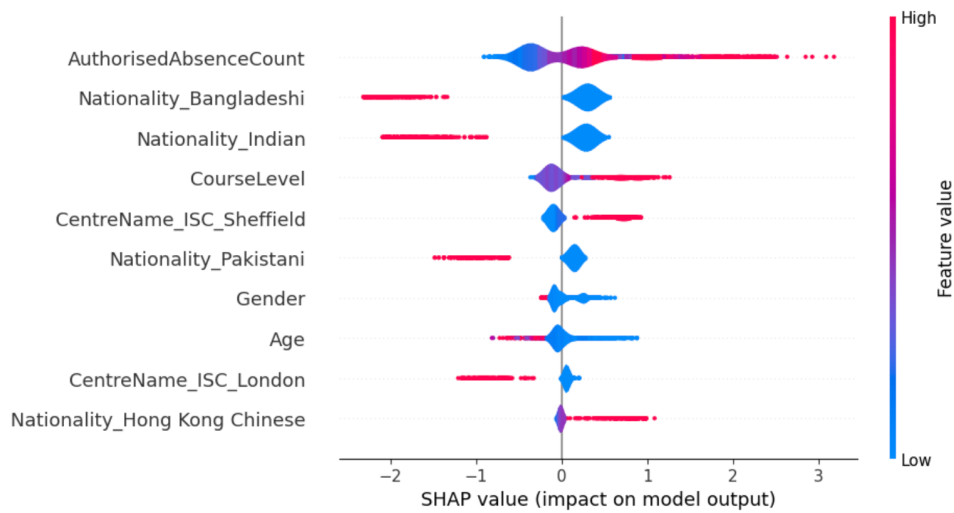


Figure 16. XGBoost Stage 2 SHAP violin chart

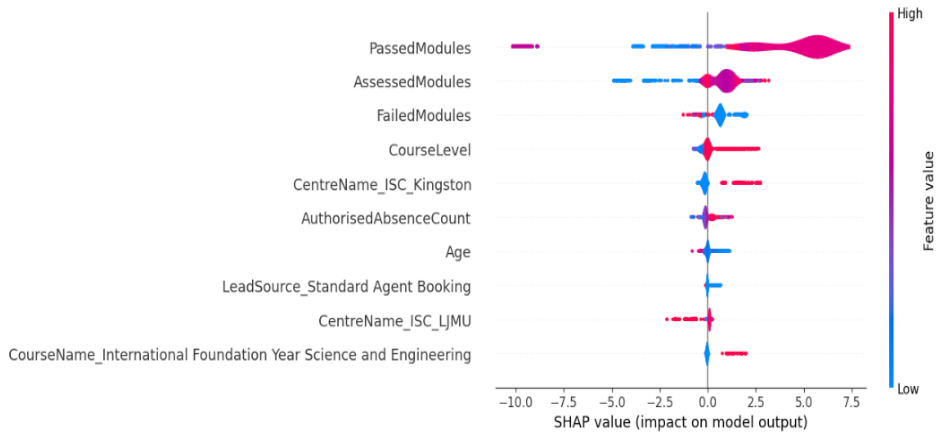


Figure 17. XGBoost Stage 3 SHAP violin chart

4.2 Implications

These findings suggest that while academic performance indicators provide the most accurate predictions, earlier-stage models remain valuable for proactive intervention. Institutions could use Stage 1 and Stage 2 models to flag students who may require additional support early in their programme, such as academic guidance or engagement initiatives. Stage 3 models can then provide more precise identification of high-risk students, enabling targeted academic support or remedial actions. Integrating predictive models across these stages could help institutions improve student retention and allocate resources more effectively.